

Data Flow Diagram

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Team ID	SWTID-2026-4683
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks
Submitted by	Dechen

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through the EduGenie system between the learner, web interface, FastAPI backend, AI modules, and AI model services. The diagram below follows the symbol legend and is based on the project architecture and API workflow described in the project document.

DFD Symbol Legend

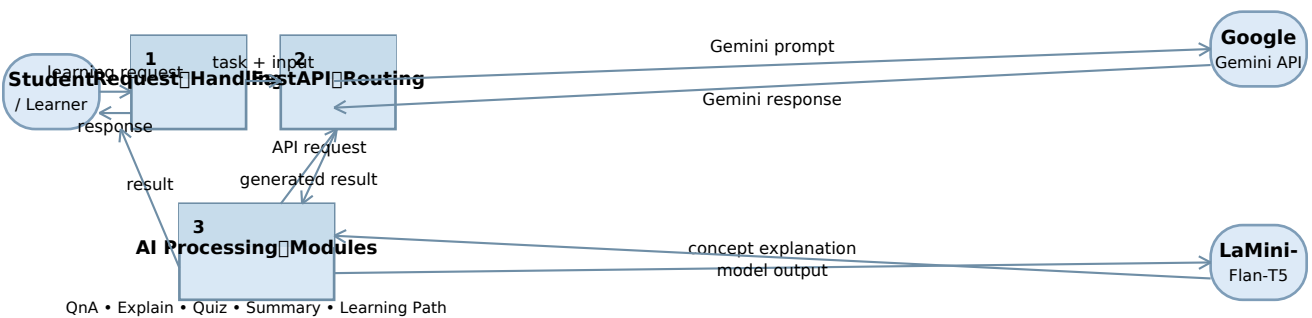
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data. Example: Student / Learner.
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data. Example: Request Handling or AI Processing.
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use. The EduGenie project document does not specify a persistent database/data store, so none is added to the main DFD.
Labelled arrow	Data Flow	The movement of data between entities, processes, and AI services, labelled with the information being passed.

Example / EduGenie DFD

The diagram below illustrates how a learner request travels through the EduGenie web interface and FastAPI backend to the appropriate AI module, with Gemini used for Q&A, quiz generation, summarization and learning recommendations, and LaMini-Flan-T5 used for concept explanation.

EduGenie - Level 0/1 Data Flow

Based on the documented FastAPI endpoints and AI modules in the project.



Documented system flow

1. The learner submits an input through the HTML/CSS web interface. 2. FastAPI routes the request to the corresponding endpoint such as /qa, /explain, /quiz, /summarize, or /learn/recommendations. 3. The relevant AI module processes the request using Gemini or LaMini-Flan-T5 and returns the generated result to the learner.

Note: The project document describes the frontend, FastAPI endpoints, Gemini 1.5 Pro, and LaMini-Flan-T5, but does not specify a persistent database. Therefore, the DFD does not invent a database or other persistent data store.